3. $A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\}$ → dari medros

$B = \{N+3, N+5, N+7, N+10, N+12, N+18\}$ → dari referral

$C = \{N+1, N+7, N+10, N+11, N+17, N+20\}$ → dari loyalty program

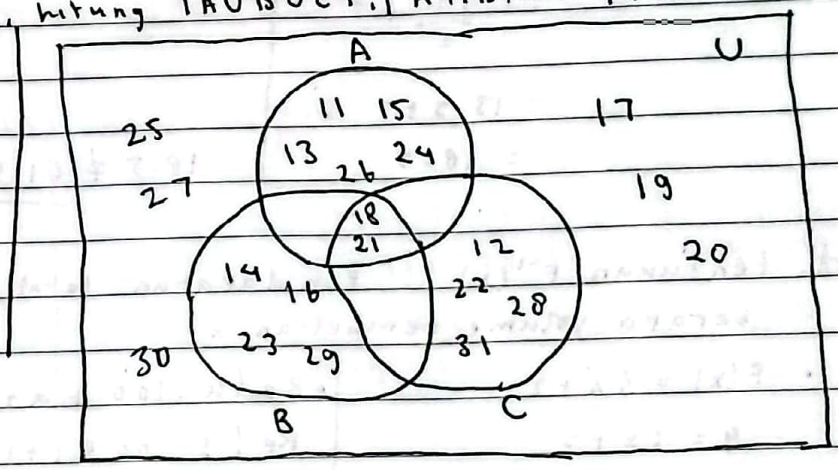
$U = \{N, N+1, N+2, \ldots N+20\}$. (total 21 pelanggan

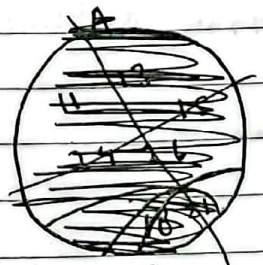
a. Gambarkan diagram venn, hitung $|A \cup B \cup C|$, $|A \cap B \cap C|$, $|(A \cup B) \cap C'|$ dan $|A \oplus B|$ $((A \cup B) - (A \cap B))$

$A = \{11, 13, 15, 18, 21, 24, 26\}$

$B = \{14, 16, 18, 21, 23, 29\}$

$C = \{12, 18, 21, 22, 28, 31\}$





- $|A \cup B \cup C| = \{11, 12, 13, 14, 15, 16, 18, 21, 22, 23, 24, 26, 28, 29, 31\}$
  $= 15$
- $|A \cap B \cap C| = \{18, 21\}$
  $= 2$
- $|(A \cup B) \cap C'| : A \cup B = \{11, 13, 14, 15, 16, 18, 21, 23, 24, 26, 29\}$
  $C = \{12, 18, 21, 22, 28, 31\}$
  $(A \cup B) \cap C' = \{11, 13, 14, 15, 16, 23, 24, 26, 29\} = 9$
- $|A \oplus B| = \{11, 13, 14, 15, 16, 18, 21, 23, 24, 26, 29\} - \{18, 21\}$
  $= \{11, 13, 14, 15, 16, 23, 24, 26, 29\} = 9$

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No.

Date :

c. $F(x) = kx + (a+b)$

$g(x) = \frac{(x^2)}{b+1} + a$

• tentukan $(f \circ g)x$ dan $(g \circ f)x$

$F(x) = 3x + 2$

$g(x) = \frac{x^2}{2} + 1$

- $(f \circ g)(x) = f(g(x))$

$= 3\left(\frac{x^2}{2} + 1\right) + 2$

$= \frac{3x^2}{2} + 3 + 2$

$(f \circ g)(x) = \frac{3x^2}{2} + 5$ //

- $(g \circ f)(x) = g(f(x))$

$= \frac{(3x+2)^2}{2} + 1$

$(3x+2)^2 = 9x^2 + 12x + 4$

$= \frac{9x^2 + 12x + 4}{2} + 1$

$(g \circ f)(x) = \frac{9x^2}{2} + 6x + 3$ //

$\boxed{\frac{3x^2}{2} + 5 \neq \frac{9x^2}{2} + 6x + 3 \text{ maka tidak komutatif}}$

• Contoh numerik $x = b + 2$

$b + 2 = 1 + 2 = 3$

- $(f \circ g)(3) = \frac{3(3^2)}{2} + 5$

$= \frac{27}{2} + 5$

$= 13.5 + 5$

$= 18.5$

- $(g \circ f)(3) = \frac{9(3^2)}{2} + 6(3) + 3$

$= \frac{81}{2} + 18 + 3$

$= 40.5 + 21 = 61.5$

$18.5 \neq 61.5$ maka terbukti tidak komutatif

d. tentukan $F^{-1}(x)$. Pendapatan total 1 pelanggan Rp $(k.100 + a + b)$ ribu berapa volume pembelian x

• $F(x) = 3x + 2$

$y = 3x + 2$

$x = 3y + 2$

$x - 2 = 3y$

$y = \frac{x-2}{3}$

$F^{-1}(x) = \frac{x-2}{3}$ //

• Rp $(k.100 + a + b)$ ribu

Rp $(3.100 + 1 + 1)$ ribu

= Rp 302 ribu //

$F(x) = 302$

• $3x + 2 = 302$

$3x = 300$

$x = 100$ //

e. klasifikasi fungsi f dan g

•) $F(x) = 3x + 2$ [redacted]

• $F(x_1) = F(x_2) \rightarrow x_1 = x_2$

misal : $3x_1 + 2 = 3x_2 + 2$

$3x_1 = 3x_2$

$x_1 = x_2 \rightarrow$ injektif

• $y \in R$, cari x

$y = 3x + 2$ [redacted]

$x = \frac{y-2}{3} \rightarrow$ surjektif //

• karena injektif dan surjektif maka f bijektif //

•) $g(x) = \frac{x^2}{2} + 1$

• misal: $g(2) = \frac{4}{2} + 1 = 3$

$g(-2) = \frac{4}{2} + 1 = 3$

Padahal $2 \neq -2$ tapi $g(2) = g(-2) \rightarrow$ tidak injektif //

• $x^2 \geq 0$ maka $\frac{x^2}{2} + 1 \geq 1$

karena kodomain = R, maka bilangan real < 1 [redacted] tidak punya pasangan $\rightarrow$ tidak surjektif

• karena tidak injektif dan surjektif maka g tidak bijektif

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